General Chemistry I

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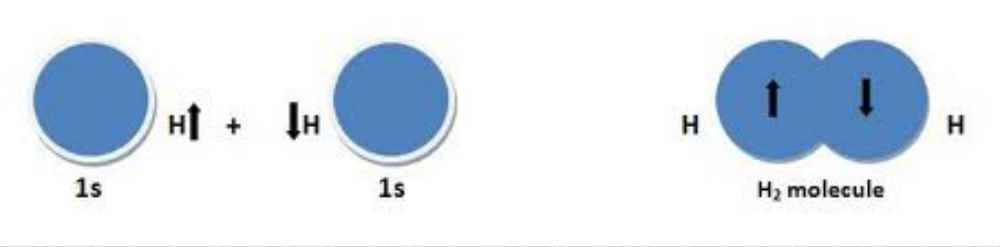
Lecture No. 9

Valence Bond Theory

9. Valence Bond Theory

Concept

- Valence Bond Theory focuses on how the atomic orbitals of the dissociated atoms combine to give individual chemical bonds when a molecule is formed.
- In other words, it considers the overlap of the atomic orbitals of the participation atoms to form a chemical bond. Due to the overlapping, electrons are localized in the bond region.



9. Valence Bond Theory

Concept

- According to the Valence Bond Theory, a bond between two atoms is formed when a pair of electrons with their spins paired is shared by two overlapping atomic orbitals, one orbital form each of the atoms joined by the bond.
- By overlap of orbitals , we simply mean that portions of the two atomic orbitals from different atoms share the same space.
- An important part of the theory is that only one pair of electrons, with their spins paired, can be shared by two overlapping orbitals.

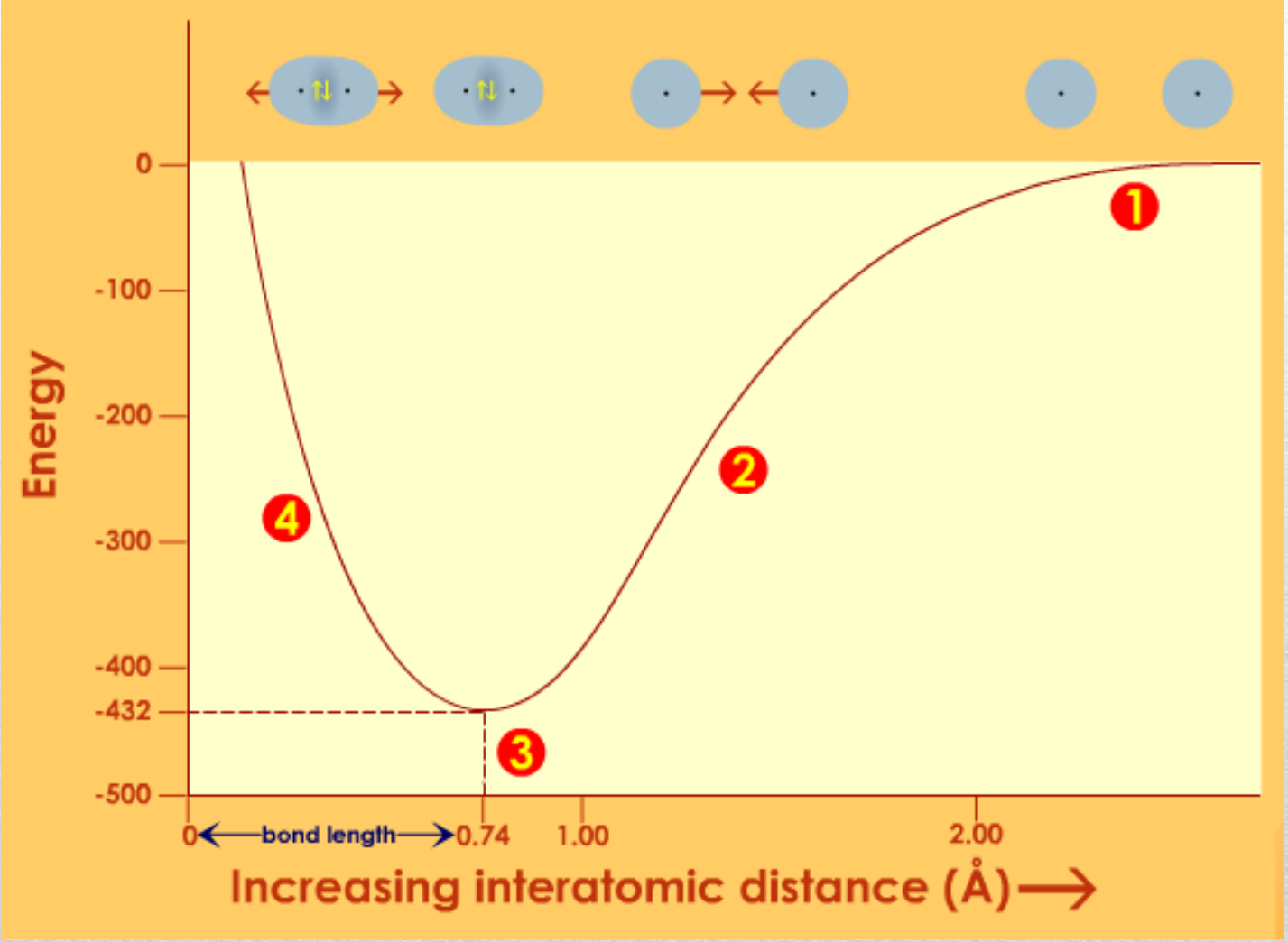
9. Valence Bond Theory

Concept

- This electron pair becomes concentrated in the region of overlap and helps “cement” the nuclei together, so the amount that the potential energy is lowered when the bond is formed is determined by the extent to which the orbitals overlap.
- Atoms tend to position themselves so that *the maximum amount of orbital overlap occurs because this yields the minimum potential energy and therefore the strongest bonds.*

9. Valence Bond Theory

Concept



9. Valence Bond Theory

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- The overlapping atomic orbitals can be of different types, for example, a sigma bond may be formed by the overlapping the following AOs.

Chemical bonds formed due to overlap of atomic orbitals					
s-s	*s-p*	*s-d*	*p-p*	*p-d*	*d-d*
H-H Li-H	H-C H-N H-F	H-Pd in palladium hydride	C-C P-P S-S	F-S in SF_6	Fe-Fe

9. Valence Bond Theory

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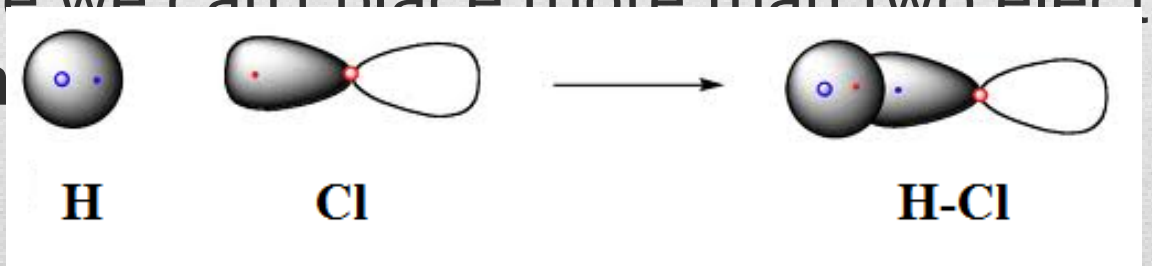
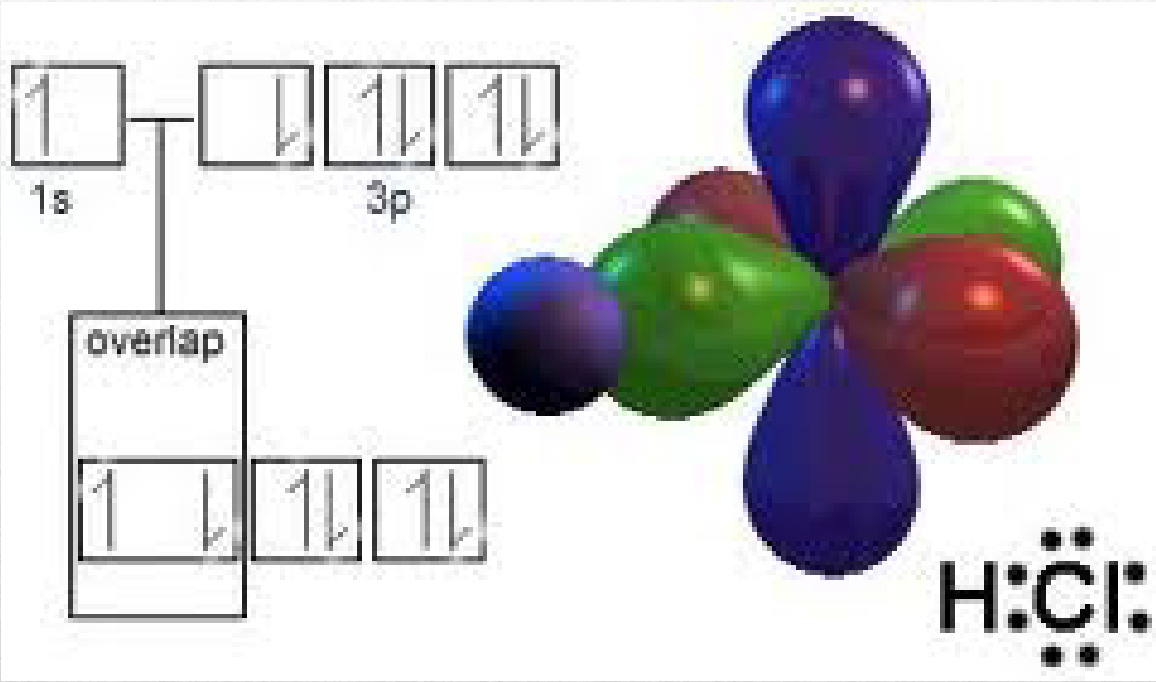
- The H-Cl bond is formed by pairing of electrons, one from hydrogen and one from chlorine. To explain this according to the VB theory, we must have two half-filled orbitals, one from each atom, that can be joined by overlap.
- Note: the bonding orbitals must be half-filled, because we can't place more than two electrons into the bon



9. Valence Bond Theory

Concept

- For HCl molecule:

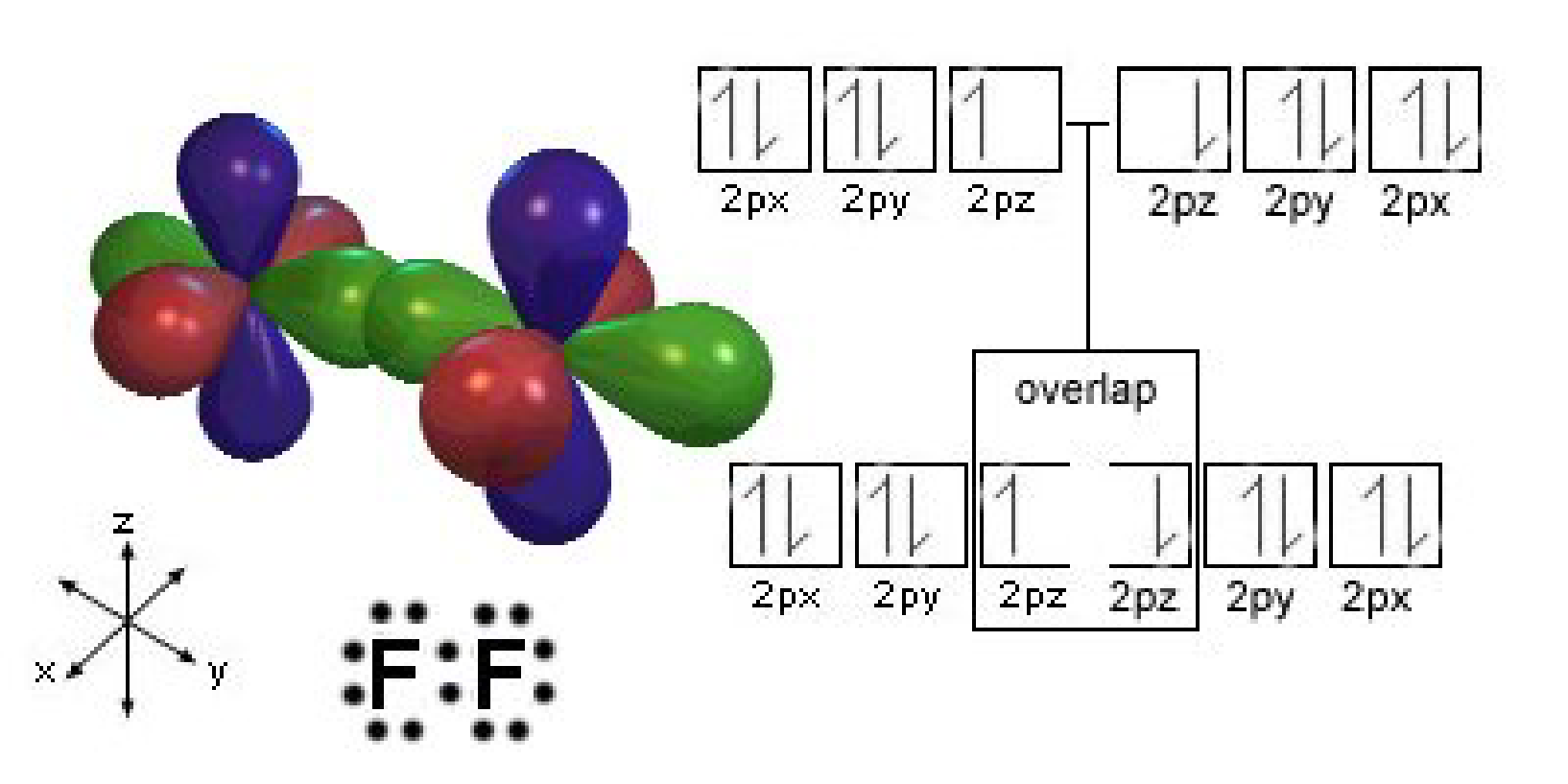


Note that the overlapping occurs when the hydrogen atom lies along the axis of the p orbital.

9. Valence Bond Theory

Concept

- For F_2 molecule:



Note the orientation of the p-orbitals

End of Lecture 9